Physics Informed Multimodal Graph Boosting Ensemble for Organic Micropollutant Rejection by Nanofiltration and Reverse Osmosis Membranes

Chirag Mishra,Jahnvi Singh

This file collects the related work and all mathematical expressions used in the main paper. Equation numbers carry the prefix S; the equations printed in the main paper correspond to Eq. (S5) (Ferry–Renkin coefficient), Eq. (S24) (blend) and Eqs. (S27) and (S28) (R^2 and RMSE).

1 Related Work

1.1 Rejection mechanisms and classical models

Trace organic micropollutants (OMPs) persist through conventional treatment and have been linked to reproductive disruption in aquatic organisms and to antibiotic resistance. Polyamide (PA) nanofiltration (NF) and reverse-osmosis (RO) membranes are key barriers for their removal. Rejection is governed jointly by steric exclusion, electrostatic interaction between the solute and the typically negatively charged PA surface, and solute–membrane affinity effects such as hydrophobic adsorption and hydrogen bonding . Classical transport models describe these mechanisms but rely on idealised assumptions about membrane homogeneity.

1.2 Machine learning for membrane rejection

Tree ensembles with SHAP explanations dominate recent work. Zhu et al. found XGBoost to be the strongest learner on a large NF/RO dataset and identified molecular volume as a key descriptor, and Wang et al.combined data-driven learning with domain knowledge, adding mechanism-inspired features such as size ratio and charge product. Xiao et al introduced multimodal fusion for this task by combining tabular inputs with molecular graphs, images or fingerprints inside the GrowNet gradient-boosting neural network.

1.3 Multimodal molecular representations

Molecular graphs preserve atomic connectivity, bond order and local environments that scalar descriptors discard. Graph isomorphism networks are among the most expressive message-passing architectures, and the GINE variant adds edge (bond) features. Structural-key fingerprints such as MACCS and calculated RDKit descriptors provide complementary, low-cost views of the same molecules.

1.4 Gradient boosting and ensembling

Gradient boosting builds an additive model by fitting each new learner to the residual error of the current ensemble . XGBoost, LightGBM and CatBoost remain the strongest general-purpose learners on medium-sized tabular data, whereas GrowNet replaces trees by shallow neural weak learners. Stacked generalisation and convex blending combine heterogeneous learners whose errors are partially uncorrelated.

1.5 Physics-informed features

Hindered-transport theory expresses the steric partitioning of a rigid sphere in a cylindrical pore through the Ferry–Renkin equation. Encoding such closed-form relations as explicit inputs is a lightweight alternative to physics-informed network architectures when data are limited.

1.6 Evaluation practice and interpretation

Single-split comparisons are noisy at small sample sizes; bootstrap resampling quantifies the resulting uncertainty and is applied in this work. Model behaviour is interpreted with Shapley additive explanations.

2 Mathematical Formulation

Throughout, i indexes records, n is the number of records in the partition under consideration, y_i is the experimental rejection (%) and $\hat{y}_i$ the predicted rejection.

2.1 Preprocessing

For each numeric feature j, missing values are replaced by the median of the observed training values,

$$x_{ij} \leftarrow m_j, \qquad m_j = \text{median}\{x_{kj} : k \in \mathcal{D}_{\text{train}},\ x_{kj} \text{ observed}\}. \tag{S1}$$

The 19 raw features entering the graph network are MinMax-scaled with training-set statistics only,

$$\tilde{x}_{ij} = \frac{x_{ij} - \min_{k\in\mathcal{D}_{\text{train}}} x_{kj}}{\max_{k\in\mathcal{D}_{\text{train}}} x_{kj} - \min_{k\in\mathcal{D}_{\text{train}}} x_{kj}}. \tag{S2}$$

2.2 Physics-informed descriptors

Let r_s be the solute radius, r_p the pore radius, $L_{\min}$ and $L_{\max}$ the minimum and maximum molecular projections, ζ the zeta potential, q the solute charge, θ the

contact angle, J_w the pure-water flux, ΔP the pressure and t the filtration duration. The thirteen descriptors are defined below.

$$\text{steric ratio:} \quad \rho = \frac{r_s}{r_p}, \qquad \text{and its square } \rho^2, \tag{S3}$$

$$\text{reduced size:} \quad \lambda = \min\left(\frac{r_s}{r_p},\, 1\right), \tag{S4}$$

$$\text{Ferry–Renkin:} \quad \Phi(\lambda) = (1-\lambda)^2\left(1 - 2.104\lambda + 2.09\lambda^3 - 0.95\lambda^5\right), \tag{S5}$$

$$\text{size to cut-off:} \quad \mathrm{MW/MWCO}, \tag{S6}$$

$$\text{aspect ratio:} \quad L_{\max}/L_{\min}, \tag{S7}$$

$$\text{cross-section:} \quad L_{\min} \times L_{\max}, \tag{S8}$$

$$\text{charge product:} \quad \zeta \times q, \tag{S9}$$

$$\text{like-charge repulsion:} \quad \mathbb{1}[\zeta < 0,\ q < 0]\,|\zeta q|, \tag{S10}$$

$$\text{hydrophobic affinity:} \quad \log D \times \theta, \tag{S11}$$

$$\text{permeability:} \quad J_w/\Delta P, \tag{S12}$$

$$\text{cumulative flux:} \quad J_w \times t, \tag{S13}$$

$$\text{speciation:} \quad \Delta\mathrm{pH} = \mathrm{pH} - \mathrm{p}K_a, \tag{S14}$$

$$\text{ionised fraction:} \quad f_{\mathrm{ion}} = \frac{1}{1 + 10^{-\,\mathrm{clip}_{[-8,8]}(\mathrm{pH}-\mathrm{p}K_a)}}. \tag{S15}$$

Eq. (S5) gives the fraction of the pore cross-section accessible to a rigid sphere; $\Phi = 0$ when $\lambda = 1$.

2.3 Edge-aware graph isomorphism network (GINE)

Each atom v with nine categorical attributes $x_{v,a}$ and each bond (u, v) with three categorical attributes $x_{uv,b}$ is embedded by summing one embedding table per attribute (hidden dimension 128),

$$h_v^{(0)} = \sum_{a=1}^{9} E_a^{\mathrm{atom}}[x_{v,a}]\,, \qquad e_{uv} = \sum_{b=1}^{3} E_b^{\mathrm{bond}}[x_{uv,b}]\,. \tag{S16}$$

Each of the two GINEConv layers updates the atom representations by

$$h_v^{(l+1)} = \mathrm{MLP}^{(l)}\Big((1+\epsilon^{(l)})\,h_v^{(l)} + \sum_{u \in \mathcal{N}(v)} \mathrm{ReLU}\big(h_u^{(l)} + e_{uv}\big)\Big), \qquad l = 0, 1, \tag{S17}$$

where $\mathcal{N}(v)$ is the set of neighbours of v and $\epsilon^{(l)}$ is a scalar. Layer normalisation and residual connections are applied around each layer as described in the main text. The graph embedding is the mean over the $|V|$ atoms of the molecule,

$$g = \frac{1}{|V|}\sum_{v \in V} h_v^{(2)}. \tag{S18}$$

In parallel, the scaled tabular vector $\tilde{x}$ is encoded by a two-layer MLP with layer normalisation and dropout (0.20), $t = \mathrm{MLP}_{\mathrm{tab}}(\tilde{x})$. The concatenated embedding

passes through the regression head (128→64→1),

$$\hat{y}^{\mathrm{GINE}} = f_{\mathrm{head}}\big([\, g \,\|\, t \,]\big), \tag{S19}$$

and the network is trained with the mean-squared-error loss over a mini-batch $\mathcal{B}$,

$$\mathcal{L} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \big(\hat{y}_i - y_i\big)^2. \tag{S20}$$

2.4 Gradient-boosted and randomised trees

Boosted tree models form an additive ensemble of M trees f_m with learning rate η [?],

$$\hat{y}_i = \sum_{m=1}^{M} \eta\, f_m(\mathbf{x}_i). \tag{S21}$$

XGBoost minimises a regularised objective, where T is the number of leaves of a tree, w its leaf weights, and α and λ_2 are the ℓ_1 and ℓ_2 penalties (0.1 and 1.5 in Table 2 of the main paper),

$$\mathcal{L}_{\mathrm{XGB}} = \sum_i (y_i - \hat{y}_i)^2 + \sum_{m=1}^{M} \Omega(f_m), \qquad \Omega(f) = \gamma T + \tfrac{1}{2}\lambda_2 \|w\|_2^2 + \alpha \|w\|_1. \tag{S22}$$

LightGBM and CatBoost use the same additive form with different tree-growing strategies. ExtraTrees averages M randomised trees,

$$\hat{y}_i = \frac{1}{M} \sum_{m=1}^{M} T_m(\mathbf{x}_i). \tag{S23}$$

2.5 Validation-selected blending

For K base predictors the blended prediction is a clipped convex combination,

$$\hat{y} = \mathrm{clip}_{[0,100]}\Big(\sum_{k=1}^{K} w_k \hat{y}_k\Big), \qquad w_k \geq 0,\ \sum_k w_k = 1. \tag{S24}$$

Single models correspond to a unit weight vector. Pairwise blends use

$$\hat{y} = \mathrm{clip}_{[0,100]}\big(w \hat{y}_a + (1 - w)\hat{y}_b\big) \tag{S25}$$

with w on a 49-point grid, and tri-model blends optimise w on the simplex with SLSQP,

$$w^\star = \arg\min_{w \in \Delta^2}\ \mathrm{RMSE}_{\mathrm{val}}\Big(\,\mathrm{clip}_{[0,100]}\big(\textstyle\sum_{k=1}^{3} w_k \hat{y}_k\big)\Big), \tag{S26}$$

where $\Delta^2 = \{w \in \mathbb{R}^3 : w \geq 0,\ \sum_k w_k = 1\}$ is the probability simplex. Among all candidates the configuration with the lowest validation RMSE is retained and applied once to the test partition. The selected weights were (0.193, 0.322, 0.486) for GINE, XGBoost and LightGBM (their sum differs from one only through rounding).

2.6 Evaluation metrics

$$R^2 = 1 - \frac{\sum_i (\hat{y}_i - y_i)^2}{\sum_i (\bar{y} - y_i)^2}, \tag{S27}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_i (\hat{y}_i - y_i)^2}, \tag{S28}$$

$$\mathrm{MAE} = \frac{1}{n}\sum_i \left|\hat{y}_i - y_i\right|, \tag{S29}$$

where $\bar{y}$ is the mean experimental rejection. The share of predictions within a tolerance τ percentage points, and the MAE within a rejection bin $\mathcal{R}$ containing $n_{\mathcal{R}}$ records, are

$$P_{\le\tau} = \frac{1}{n}\sum_i \mathbb{1}\big[\,|\hat{y}_i - y_i| \le \tau\,\big], \qquad \mathrm{MAE}_{\mathcal{R}} = \frac{1}{n_{\mathcal{R}}}\sum_{i\in\mathcal{R}} |\hat{y}_i - y_i|\,. \tag{S30}$$

2.7 Bootstrap confidence intervals and paired comparisons

With $B = 5000$ resamples, for each $b = 1, \ldots, B$ draw indices $i_1^{(b)}, \ldots, i_n^{(b)}$ uniformly with replacement from the $n = 162$ test records and evaluate a metric $\mathcal{M}^{(b)}$ (for example R^2) on the resampled set. The 95% percentile interval is

$$\big[\, q_{0.025}\{\mathcal{M}^{(b)}\},\ q_{0.975}\{\mathcal{M}^{(b)}\}\,\big]. \tag{S31}$$

For two models evaluated on the same resamples, the paired difference, its mean, and the probability that the blend is better are

$$\Delta^{(b)} = {R^2_{\mathrm{blend}}}^{(b)} - {R^2_{\mathrm{other}}}^{(b)}, \qquad \bar{\Delta} = \frac{1}{B}\sum_{b=1}^{B}\Delta^{(b)}, \tag{S32}$$

$$P(\text{blend better}) = \frac{1}{B}\sum_{b=1}^{B}\mathbb{1}\big[\Delta^{(b)} > 0\big]. \tag{S33}$$

The probability that the blend exceeds a reference value c (here $c = 0.9014$ for R^2) is $P(R^2 > c) = \frac{1}{B}\sum_b \mathbb{1}[{R^2}^{(b)}_{\mathrm{blend}} > c]$, and the analogous quantity for RMSE with $c = 9.1118$ uses $\mathbb{1}[\mathrm{RMSE}^{(b)}_{\mathrm{blend}} < c]$.

2.8 SHAP attribution

For a model f with feature set F, the Shapley value of feature j for a prediction at $\mathbf{x}$ is

$$\phi_j = \sum_{S\subseteq F\setminus\{j\}} \frac{|S|!\,(|F| - |S| - 1)!}{|F|!}\Big[f_{S\cup\{j\}}(\mathbf{x}) - f_S(\mathbf{x})\Big], \tag{S34}$$

and the prediction decomposes additively as

$$f(\mathbf{x}) = \phi_0 + \sum_{j \in F} \phi_j. \tag{S35}$$

TreeExplainer evaluates these values exactly for tree ensembles. The global importance of feature j over n records, and the summed importance of a feature family G (Fig. 9b of the main paper), are

$$I_j = \frac{1}{n} \sum_{i=1}^{n} |\phi_{ij}|, \qquad I_G = \sum_{j \in G} I_j. \tag{S36}$$